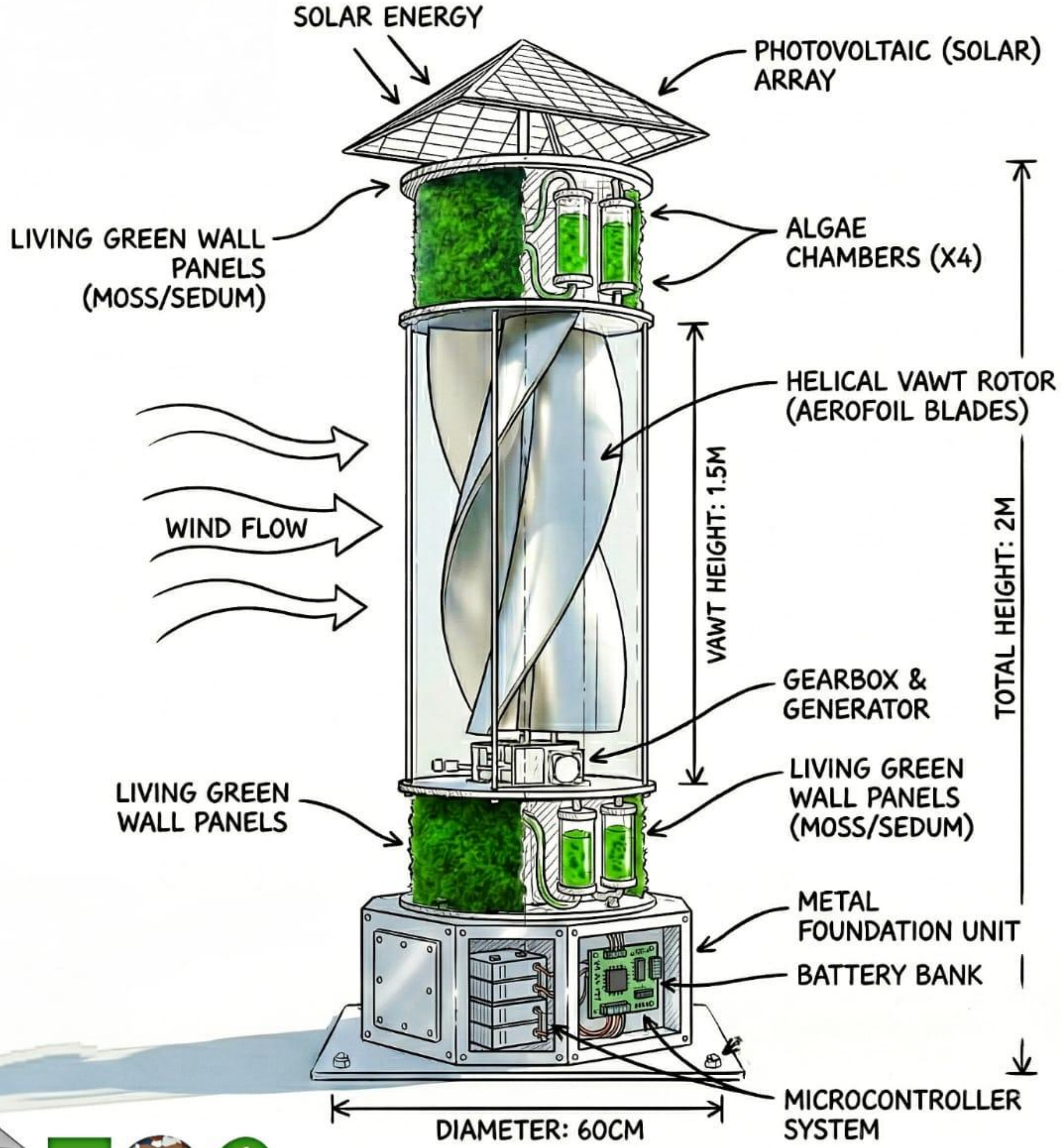
Energy Harvesting and Air Purification Solution on Highways

"Energy wasted today is a darkness we draft for tomorrow."



ECO WAY
CLEAN ENERGY SOLUTIONS

Tetranix
Engineering Intelligent Infrastructure

Project Overview: EcoWay

EcoWay is an innovative, self-sustaining roadside unit designed to transform busy highways into hubs of clean energy and air purification. It addresses two major environmental challenges: capturing unused wind energy from moving vehicles and reducing harmful air pollution.

Core Components

The system integrates three primary technologies into one compact structure:

Helical Savonius Wind Turbine (VAWT): A vertical-axis turbine with a 180° twist. Unlike traditional turbines, it captures wind from any direction and is optimized to start rotating even at low wind speeds generated by passing traffic.
Pyramid Solar Panels: A multi-faceted solar array on top of the unit. Its pyramid shape allows it to capture sunlight from different angles throughout the day, ensuring a steady power supply even when there is no wind.
Algae Biofiltration System: An active chamber containing microalgae (such as Chlorella vulgaris). Fans draw polluted air into the chamber, where the algae use photosynthesis to absorb CO_{2} and other pollutants, releasing purified oxygen back into the atmosphere.

How It Works

The project operates through a continuous cycle of harvesting and purification:

Energy Capture: The turbine and solar panels generate electricity, which is stabilized by a charge controller.
Storage: Surplus energy is stored in a Lithium-ion battery managed by a Battery Management System (BMS) for safety and 24/7 operation.
Active Purification: The stored energy powers internal fans that actively pump highway air into the algae chambers for cleaning.
Monitoring: A microcontroller (Arduino-based) and sensors track air quality and system health in real-time.

Part	Description	Rationale/Function
Helical Savonius Blade	The 180 twisted rotor.	Captures turbulent wind with high starting torque; the twist ensures smooth torque output (CFD validated).
Central Rotor Shaft	The vertical axis supports the blades.	Transfer rotational torque from the blades to the generator located in the base casing.
Pyramid Solar Panel Housing	The multi-faceted top structure.	Houses the custom-cut Monocrystalline modules (not visible, but integrated) to maximize solar energy capture throughout the day.
Top/Bottom End Plates	Circular plates are at the top and bottom of the rotor.	Crucial for preventing spanwise spillage of wind and enhancing the pressure difference between the concave and convex sides of the blades, improving CP
Algae Chamber (Top & Bottom)	The sealed inner tanks contain the microalgae culture.	The Reactor, where biological purification occurs; microalgae break down CO_2 and NO_X.
Perforated Mesh Air Intake	The outer perforated aluminum cylinder.	Serves as the inlet filter (pre-treatment) and the central passage for drawing polluted air into the active filtration chamber.
Active Fans (Internal/Not Labeled)	DC Axial Fans (Intake and Exhaust).	Creates the active suction and circulation necessary to maintain the calculated Clean Air Delivery Rate (CADR).
Base Battery & Control Casing	The lowest, largest cylindrical enclosure.	Houses the generator, Li-ion battery pack, BMS, and charge controller. Provides structural support for the entire unit and acts as the protected hub for all electronics.
Generator (Internal/Not Labeled)	Permanent Magnet Generator (PMG).	Converts the shaft's rotation (torque) into electricity for storage.
Bearings (Internal/Not Labeled)	High-quality bearings (e.g., Magnetic or deep groove).	Allow the central shaft to rotate with minimal friction to ensure smooth operation and longevity.

Component / Parameter	Dimension	Units	Details & Justification
I. Overall System Architecture (Per Unit)			
Base Casing Height	40	cm	Main control/storage unit height.
Top Algae Chamber Mesh Height	40	cm	Height of the mesh structure surrounding the chambers.
Bottom Algae Chamber Mesh Height	40	cm	Height of the mesh structure surrounding the chambers.
Pyramidal Solar Panel Height	30	cm	Height of the solar array structure.
Turbine Blade Height	100	cm	Height of the turbine blades.
Turbine Diameter	60	cm	Matches the 60cm base constraint.
Base Casing Diameter	60	cm	Fixed diameter of the central structure.
Estimated Total Height with plates (Approximately)	260	cm	Total height is a stack of components.
II. Algae Chamber (Photobioreactor) – Stacked (4 chambers per unit x 2 units = 8 total)			
Chamber Outer Diameter	22	cm	Maximum size to fit two chambers sideby-side within the 60cm base.
Chamber Inner Diameter	21	cm	Usable diameter for culture (accounting for 0.5cm wall thickness).
Volume per Chamber	12.13	Liters	Calculated volume
Total Culture Volume (8 chambers)	97.04	Liters	Nearly doubles the previous volume for the same base footprint.

I. Core Capabilities and Validation

Parameter	Specification	CFD/FEA Validation Status
Energy Autonomy	**24/7 Continuous Operation**	**Confirmed.** Energy Balance (E_{Gen}) provides a net surplus (e.g., ~763 Wh/day).
Structural Integrity	**High FOS** (Factor of Safety)	**Confirmed.** FEA validated structural resistance to 40 N drag force; max stress is only 7.345 MPa (minimal deflection 2.884 mm).
Wind Performance	**Smooth Torque Output**	**Confirmed.** CFD Flow Trajectories confirm the 180° helical twist eliminates torque ripple and maximizes output at low, turbulent RPM.
Primary Function	**Biofiltration** (CO_2, NO_x, $PM_{2.5}$)	The system is designed to achieve maximum pollutant-algae contact time.
Design Type	**Hybrid Renewable Energy System (HRES)**	**Confirmed** optimal configuration using dual intermittent sources and battery storage.

II. Technical Specifications (Hardware & Power)

A. Energy Harvesting and Conversion

Component	Technology	Technical Specification	Rationale for Selection
Wind Turbine	**Helical Savonius VAWT**	60 cm Diameter, 100cm Height, Cp validated by CFD.	High starting torque and stability in turbulent, lowspeed roadside wind.
Solar Array	**Pyramid Monocrystalline PV**	0.75 m^2 surface area; Custom-cut triangular/trapezoidal modules.	**Highest efficiency** (18% in a compact footprint, multi-faceted mounting ensures maximum daily capture.
Generator	**Permanent Magnet DC Generator (PMG)**	Low-RPM rated (e.g., 150W to 300W	Chosen for high efficiency at the low, variable RPM of the Savonius turbine, maximizing energy harvest.
Structure Material	**Aluminum/Steel**	FEA-validated for 15 m/s wind gusts.	Optimal balance of strength, lightweight structure, and superior corrosion resistance for roadside assets.

B. Power Management and System Loads

Component	Specification	Role	Capacity Justification
Battery Storage	12 V / 25 Ah Li-ion Pack	Energy Storage for 24/7 autonomy.	Sized to ensure a 48-hour backup reserve for the continuous 5.85W system load.
Charge Controller	MPPT (Maximum Power Point Tracking)	Optimizes power transfer from both the fluctuating wind and solar sources to the battery.	Essential for maximizing total energy harvest (E_{Gen}) and system reliability.
Active Load	8 DC Axial Fans	5.85 W total continuous load.	Ensures calculated Clean Air Delivery Rate (CADR) and optimal contact time for biofiltration.

III. Environmental and Economic Value

Metric	Value/Assessment	Impact and Justification
Environmental Value	CO_2, NO_X Absorption	Provides a zero-carbon, nature-based solution to local air pollution. Uses microalgae biofiltration.
Cost (BOM)	LKR 83,050 (Approx. per unit)	The cost is currently high due to custom parts (Aluminum fabrication, custom solar modules).
Economic Viability	Simple ROI 8.19 Years	Acceptable payback period for high-quality, long-life infrastructure. Requires mass production scale-up to reduce initial unit cost and achieve competitive ROI.
Maintenance	Low	Requires service support at longer intervals due to robust design (Li-ion/BMS, no moving mechanical filters).